Attendance QR code:

Texts that have been used to generate this lecture content

- Molecular Cell Biology Lodish et al
- Molecular Biology of the Cell Alberts et al
- Principles of Biochemistry Horton et al
- Molecular Biology of the Gene Watson et al

BIOC0001

Lecture 4

Dr. Suzanne Ruddy

Gene Expression

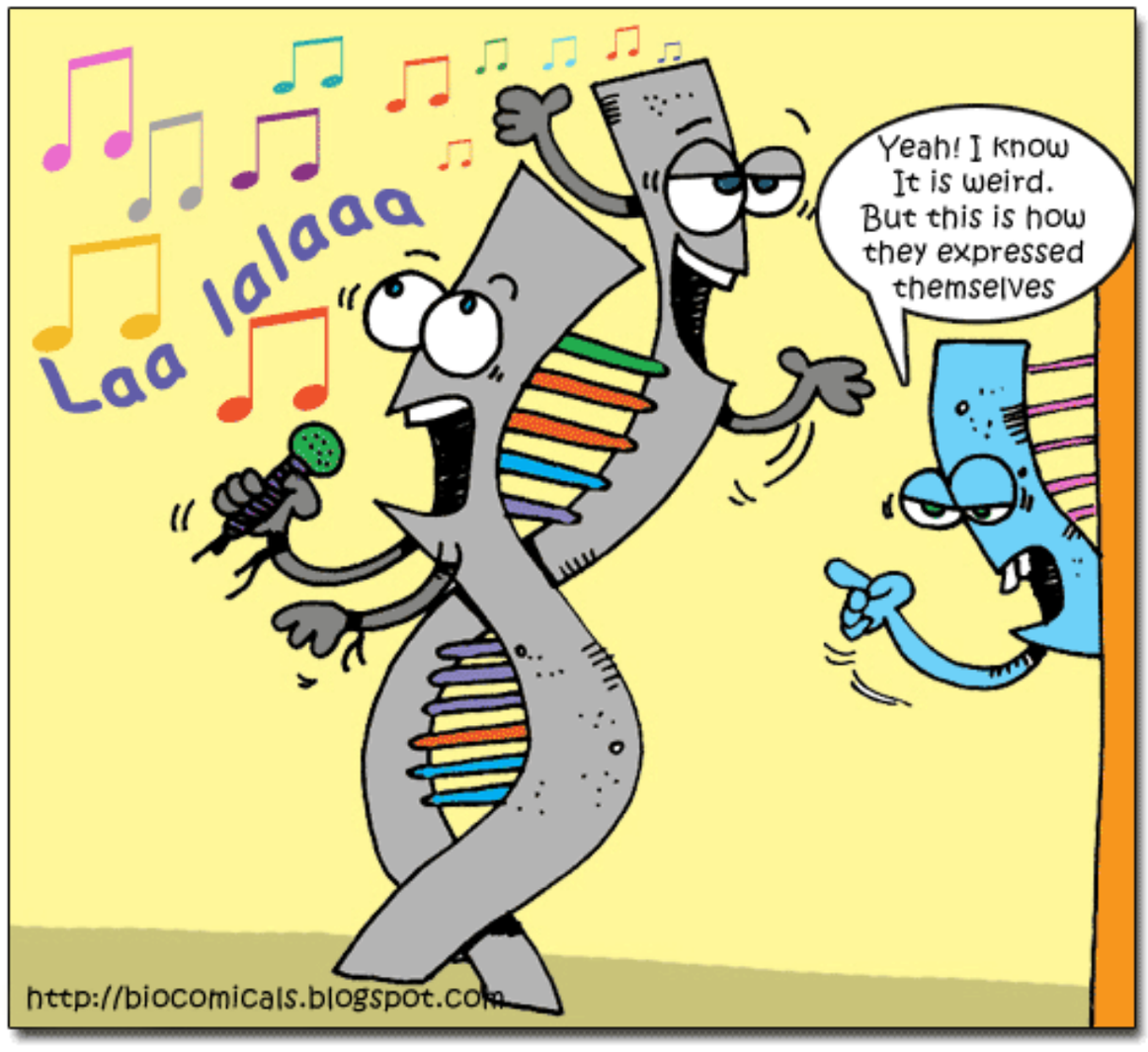
Types of RNA

Gene structure

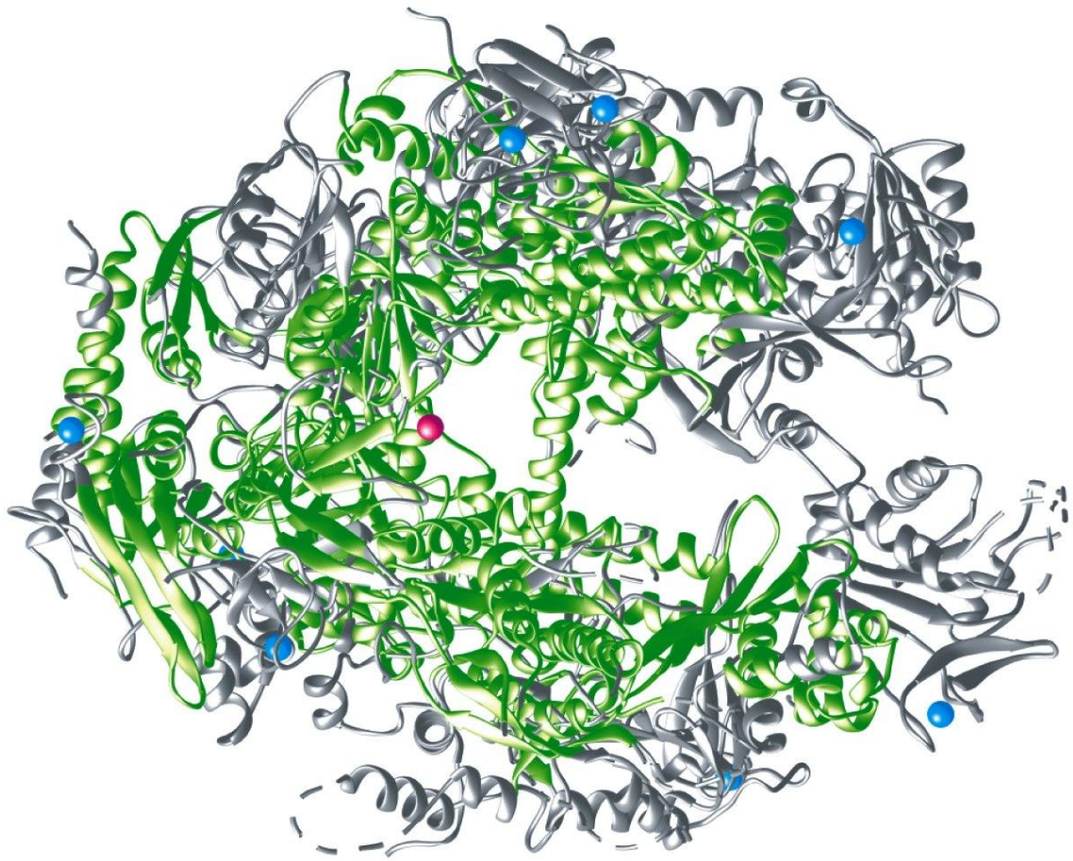
Transcriptional regulation

mRNA processing

-prokaryotes
-eukaryotes



Central Dogma



RNA Polymerase

DNA

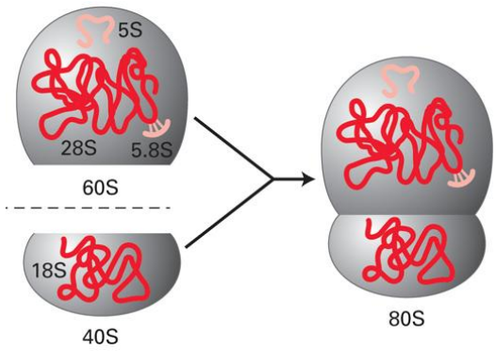
Transcription

RNA

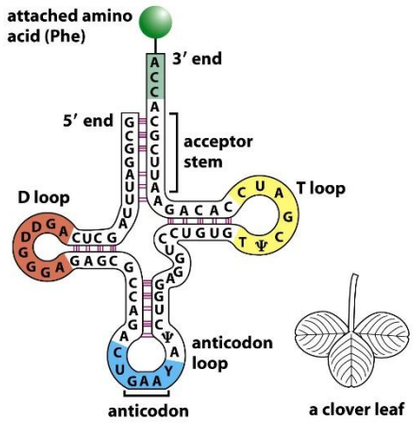
Translation

PROTEIN

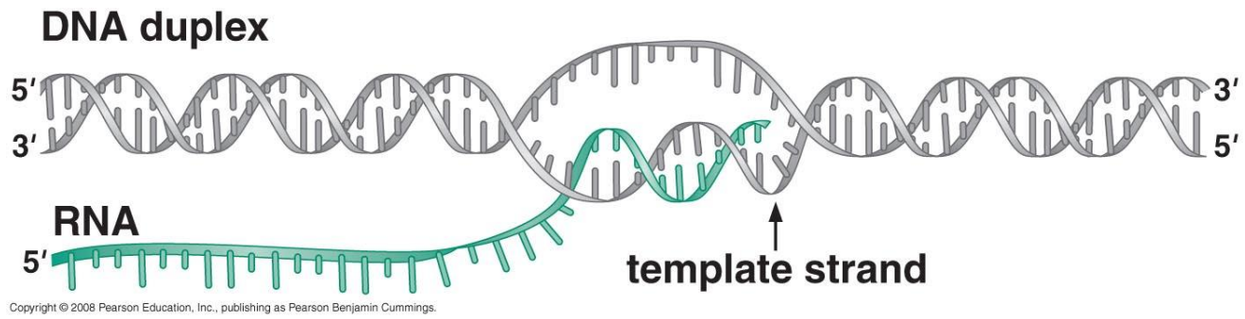
Different Types of RNA



Ribosomal RNA



Transfer RNA



Messenger RNA

TABLE 7-2 Classes of RNA Transcribed by the Three Eukaryotic Nuclear RNA Polymerases and Their Functions

POLYMERASE	RNA TRANSCRIBED	RNA FUNCTION
RNA polymerase I	Pre r-RNA (28S, 18S, 5.8S rRNAs)	Ribosome components, protein synthesis
RNA polymerase II	mRNA snRNAs miRNAs	Encodes protein RNA Splicing Post-transcriptional gene control
RNA polymerase III	tRNAs 5S rRNA snRNA U6 7S RNA Other stable short RNAs	Protein synthesis Ribosome component, protein synthesis RNA Splicing Signal-recognition particle for insertion of polypeptides into the endoplasmic reticulum Various functions, unknown for many

Table 7-2
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Coding strand (read by RNA polymerase II)

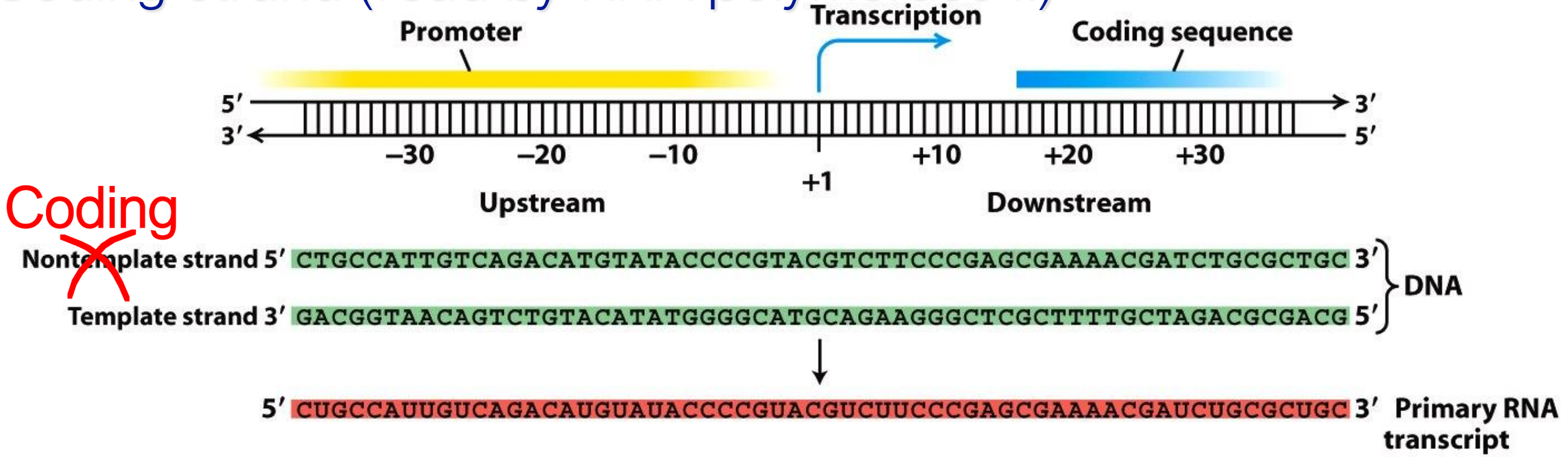
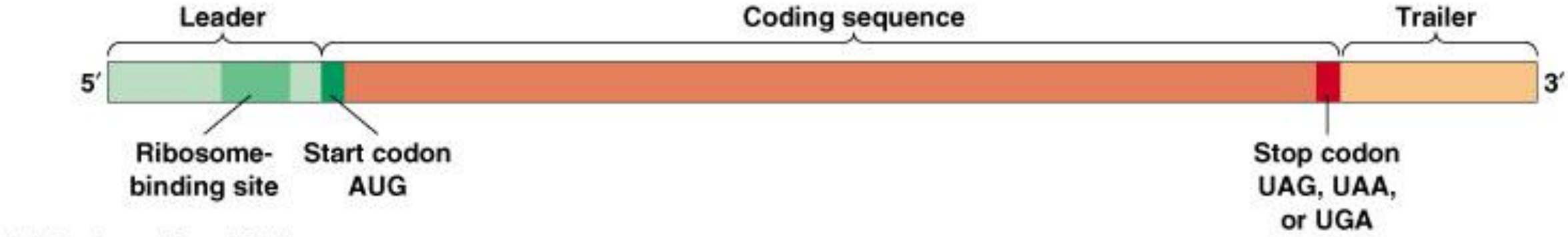
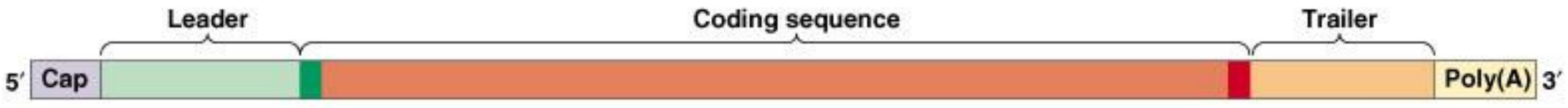


Figure 4-10b
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(a) Prokaryotic mRNA

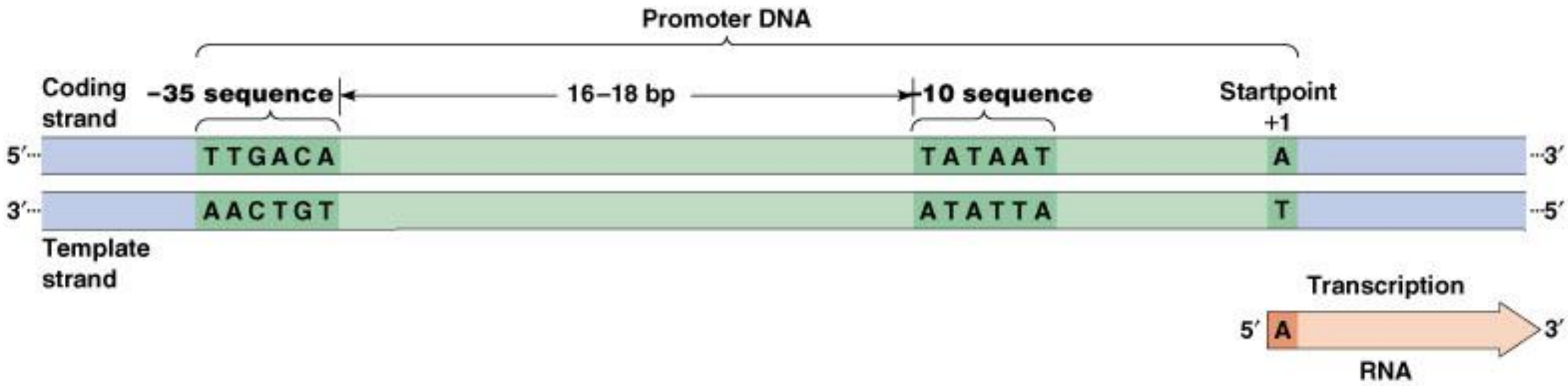


(b) Eukaryotic mRNA

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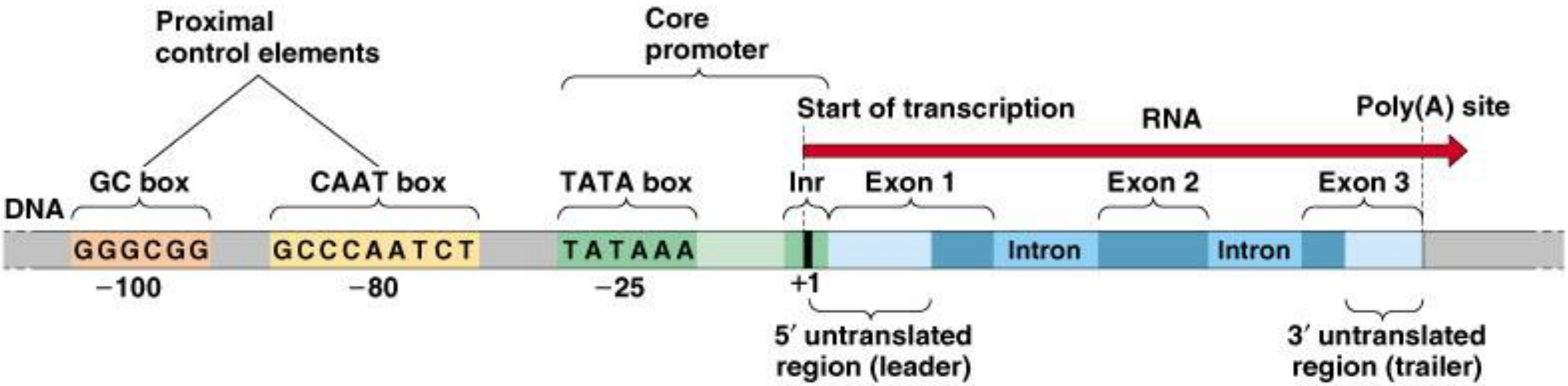
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Prokaryotic gene structure-promoters



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Eukaryotic Gene Structure- Promoters



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Table 21-3 Some Eukaryotic DNA Control Elements and Transcription Factors That Bind Them

DNA Control Element	Genes Where Found	DNA Element Consensus Sequence*	Transcription Factor**
TATA box	Many	TATAAAA (in promoter)	TFIID
CAAT box	Many	GGCCAATCT	CTF
GC box	Many	GGGCGG	Sp1
Octamer	Many	ATTTGCAT	Oct-1/Oct-2
Heat shock element	Heat shock genes	CnnGAAnnTTCnnG	Heat-shock transcription factor
Estrogen response element	Ovalbumin, others	AGGTCAnnnTGACCT	Estrogen receptor
κB	Immunoglobulin light chain κ	GGGGACTTTCC	NF-κB

*Some researchers give slightly different consensus sequences. A lowercase n signifies that any nucleotide can be located at that position.

**In some cases, other names have also been assigned to the factors listed here.

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Gene structure: protein-encoding genes

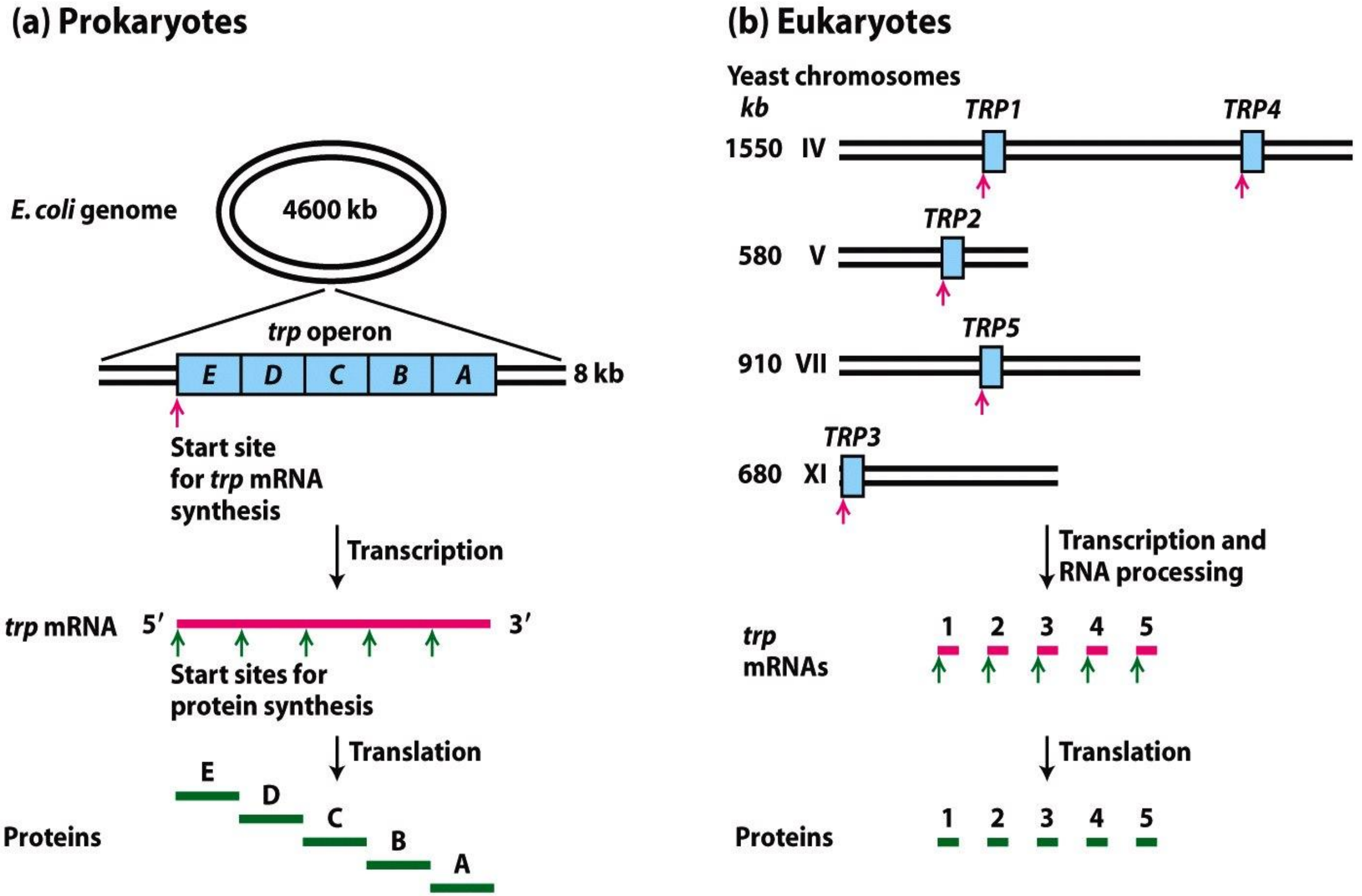


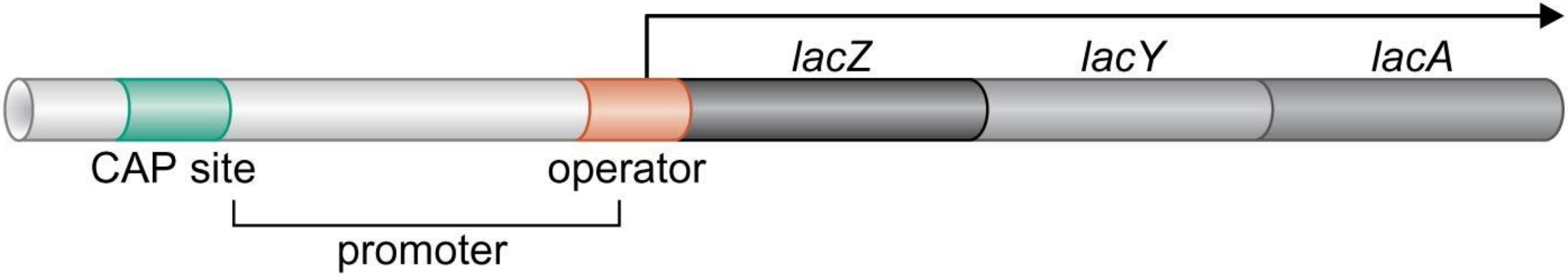
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Regulation of transcription: Prokaryotes

Prokaryotes regulate transcription as on/off or up/down

The Lac Operon (*E. Coli*)



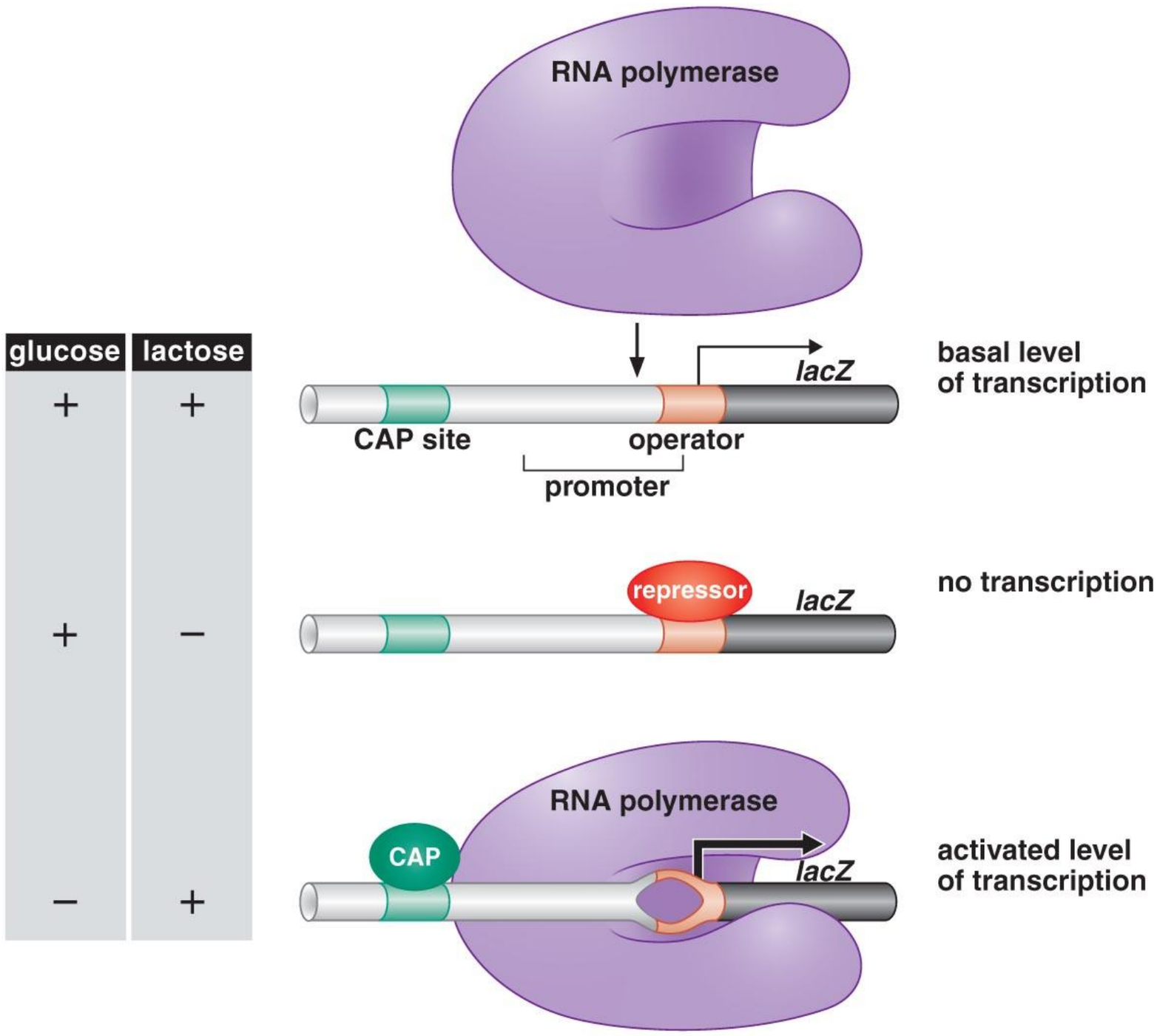
- *Lac-Z*-encodes beta galactosidase, which catalyses cleavage of lactose to glucose (the preferred substrate) and galactose.

- *Lac-Y* encodes lactose permease which inserts into the cell membrane and transports lactose across.

- *Lac-A* encodes thiogalactoside transacetylase which rids the cell of toxic thiogalactosides which are also transported across.

- Lac-I is not part of the Lac operon and encodes the repressor molecule which regulates operon transcription by binding to the operator.

Prokaryotic Regulation example: the lactose operon- E.coli



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Regulation of the lactose operon-molecular perspective

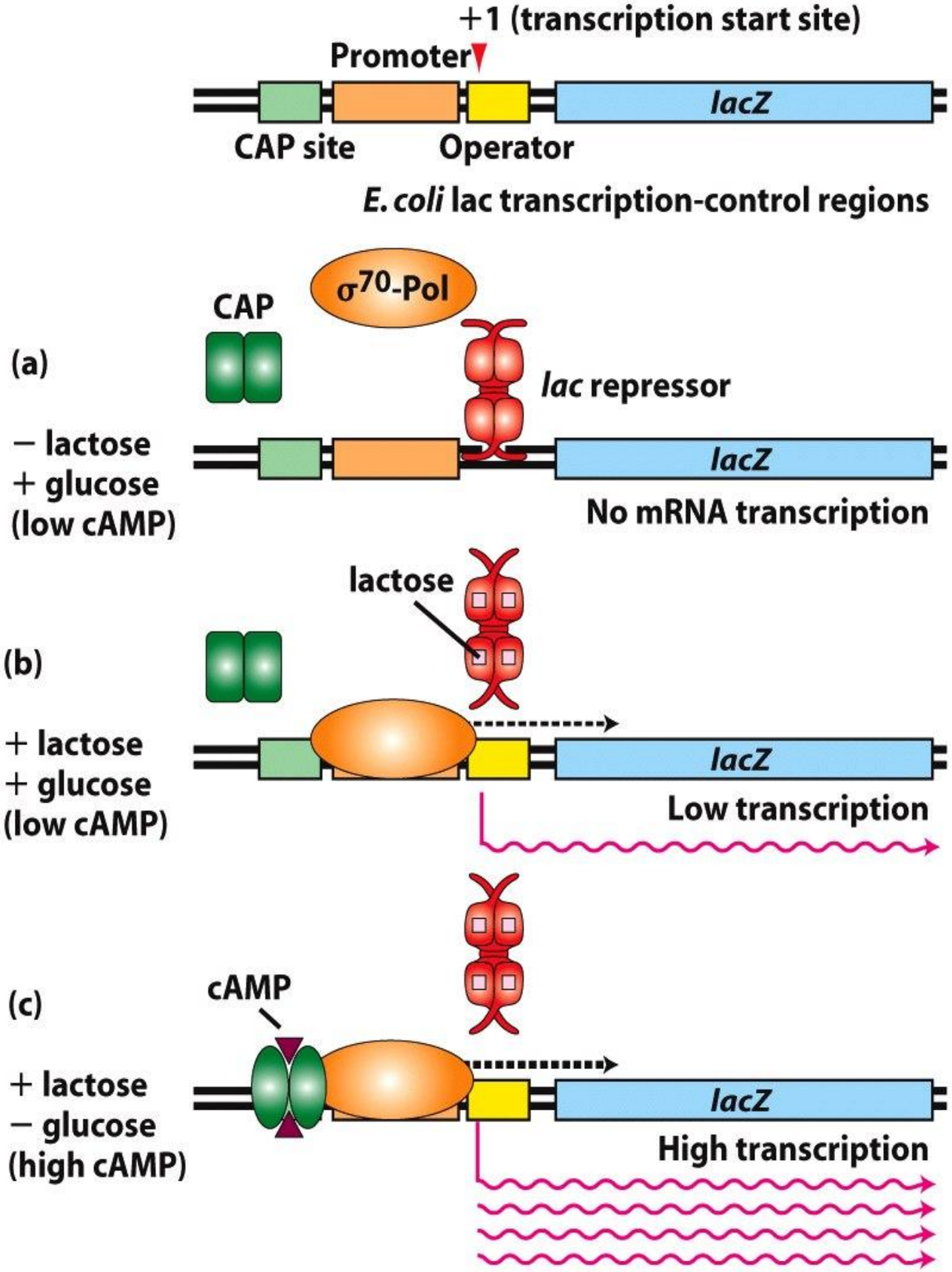
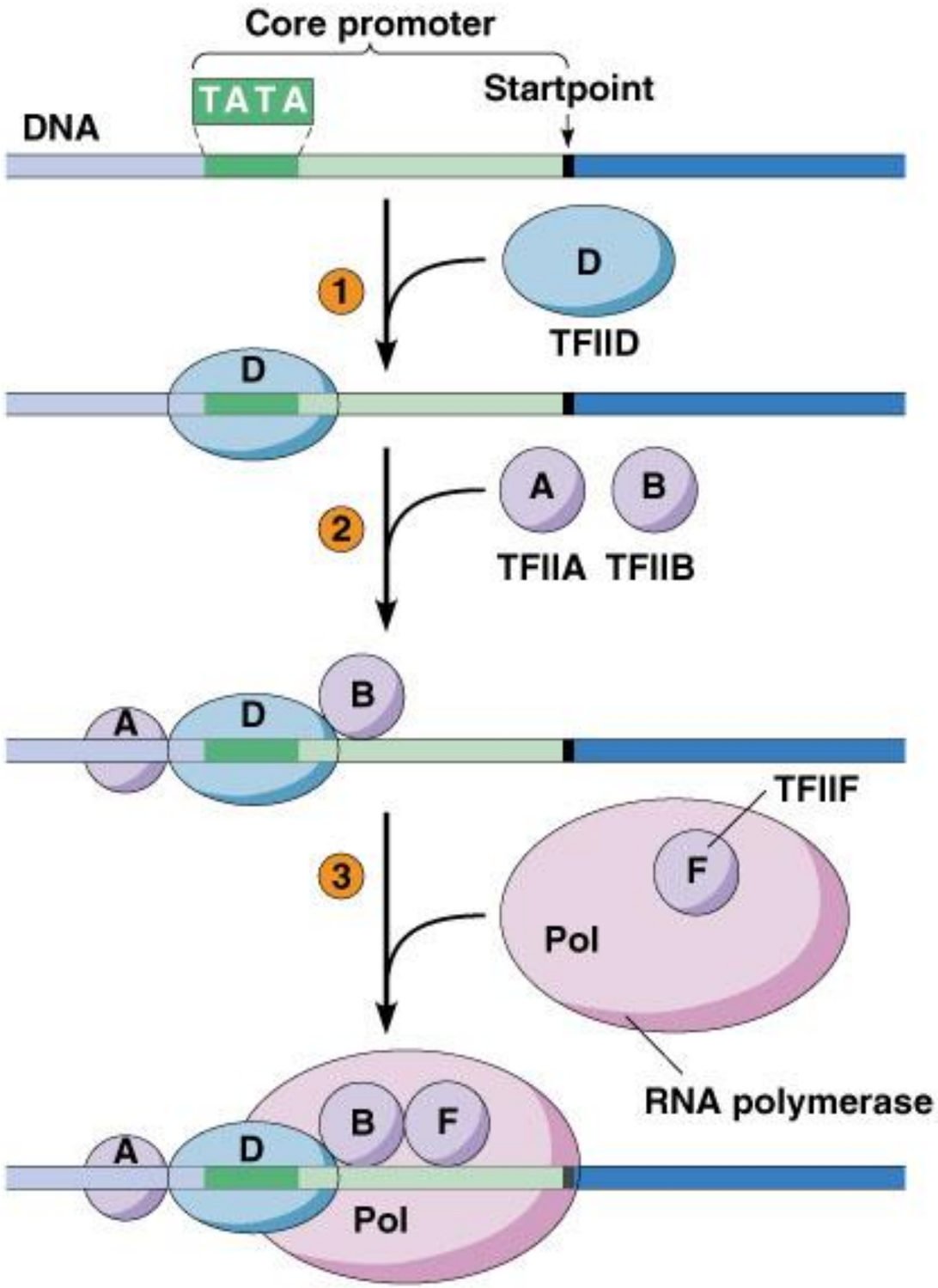


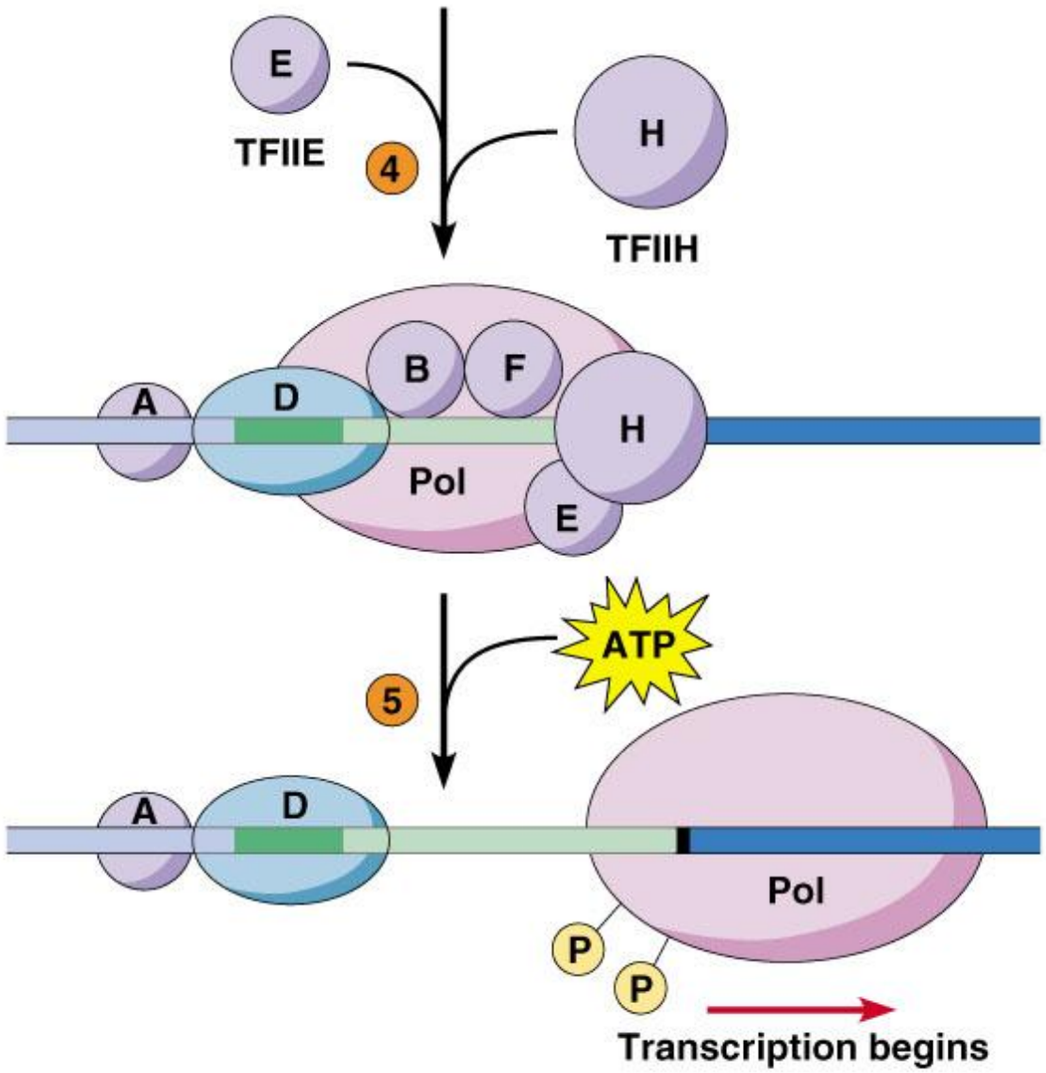
Figure 7-2
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Lactose spontaneously forms an isomer, allolactose, so that this is present at a low level when lactose is present.

The allolactose binds the repressor and alters its conformation so that it no longer binds the operator.

Eukaryotic Gene Regulation: General eukaryotic transcription factors





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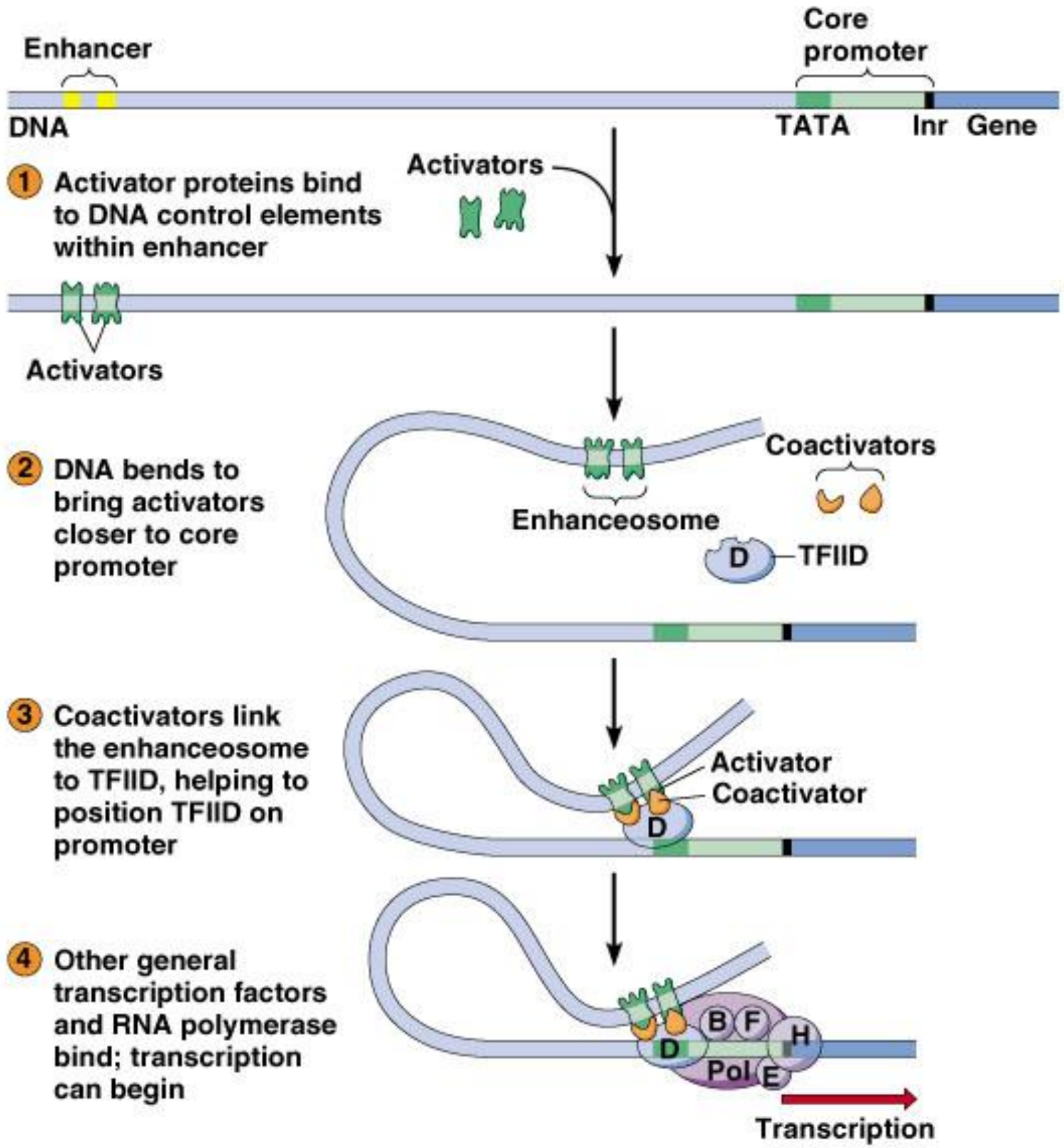
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Eukaryotic gene regulation: Additional transcription factors

Enhancer is DNA

Activators are DNA binding proteins

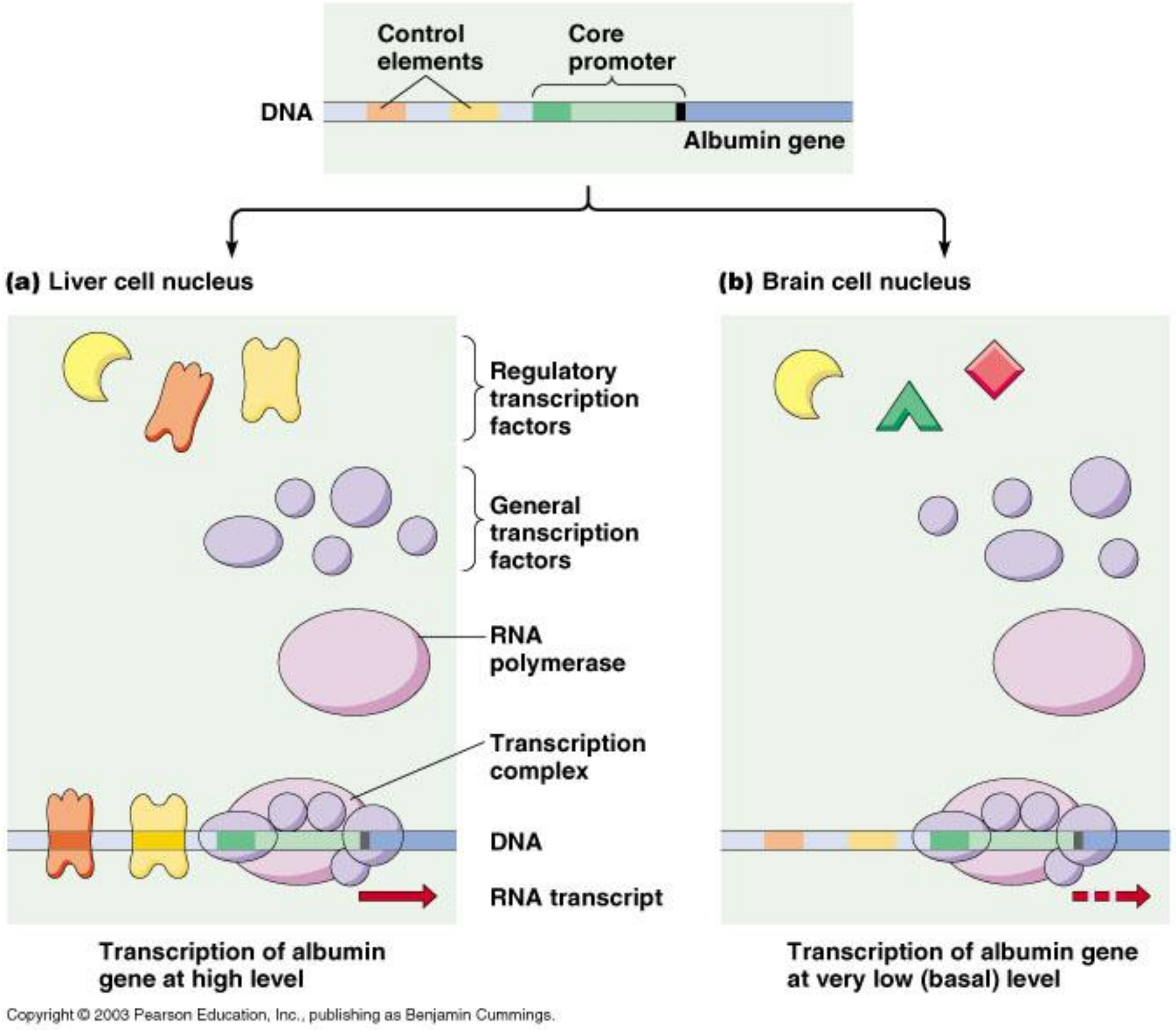


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Tissue-specific gene expression

Activators (DNA binding proteins) are tissue specific



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Making the final messenger RNA

3 key processes:

- Capping
- Splicing
- Polyadenylation

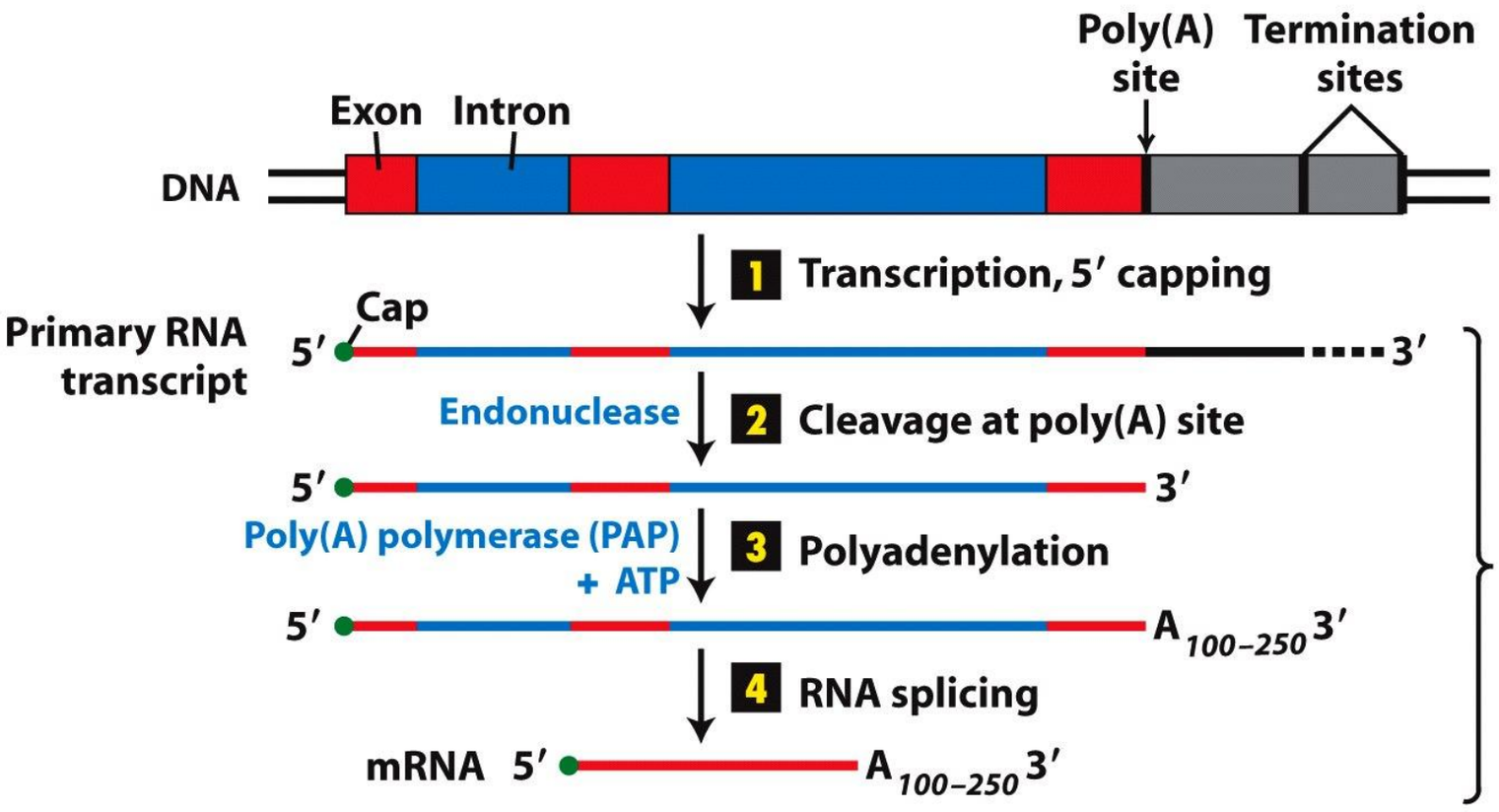
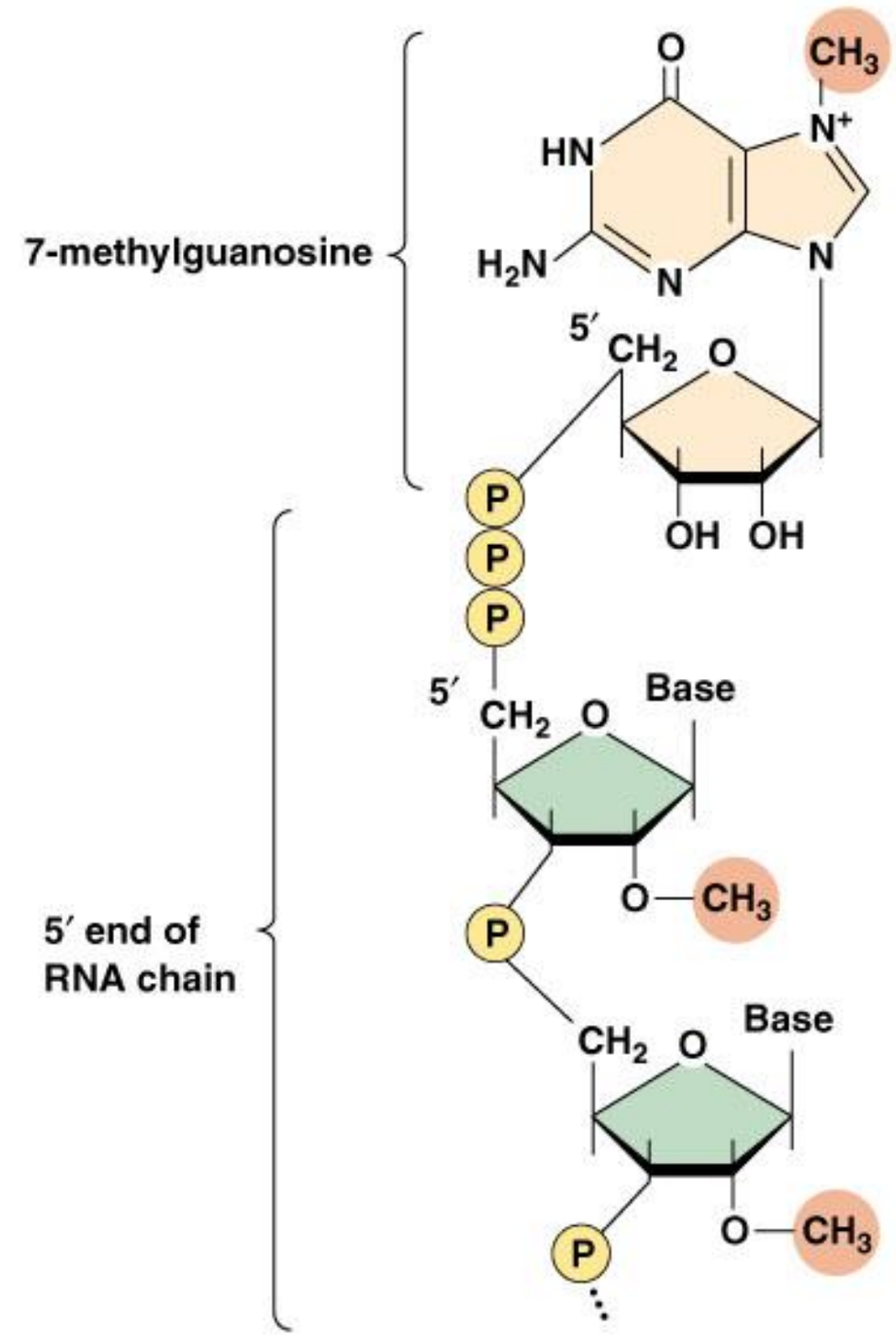


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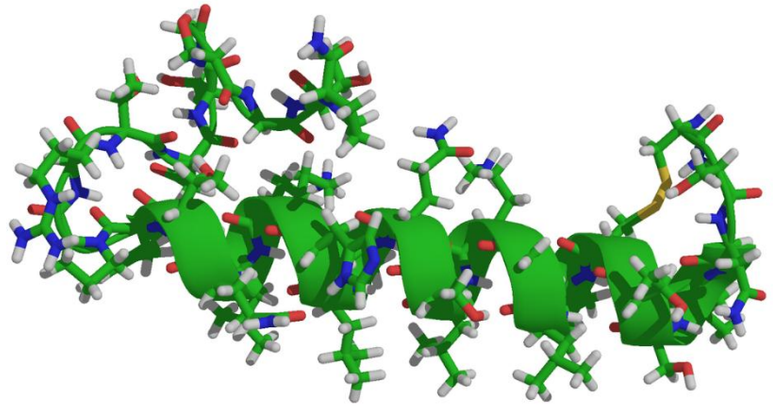


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Calcitonin (CT) and Calcitonin gene-related peptide (CGRP)

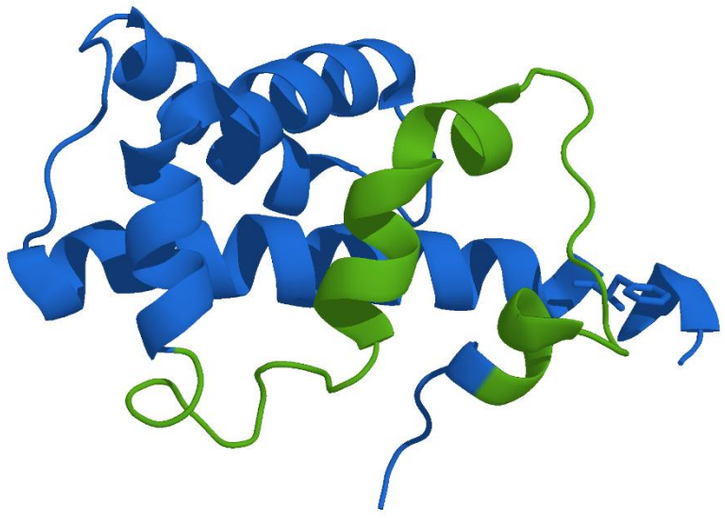
- Two peptide hormones from same gene:
- calcitonin (CT)
- calcitonin gene-related peptide (CGRP).



Calcitonin

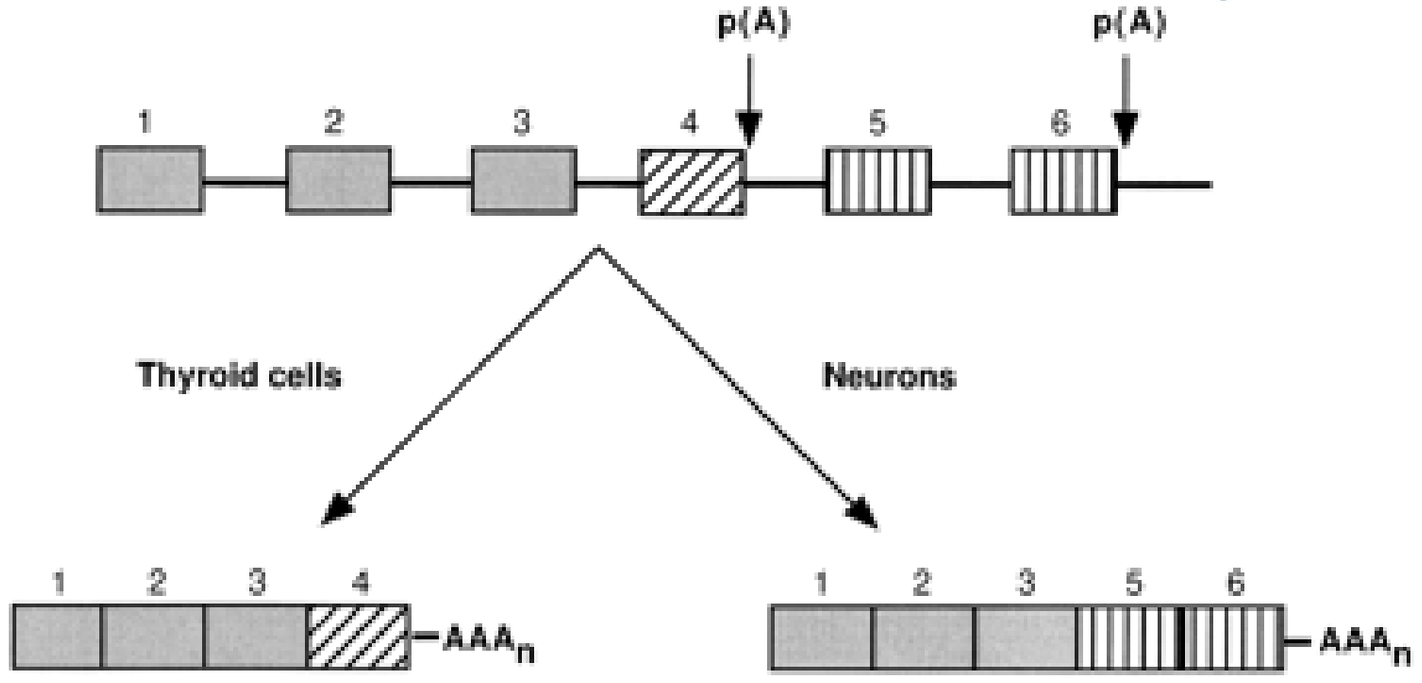
- CT (a 32-amino-acid peptide) is a hormone primarily involved in regulating calcium levels in the blood. It is expressed in the thyroid gland.

- CGRP (a 37-amino-acid neuropeptide) is the most potent endogenous vasodilatory peptide and is associated with and released during migraine. It is expressed in the brain.



Calcitonin gene related peptide

Tissue-specific RNA processing



thyroid

calcitonin

brain

CGRP

- Alternative splicing involves use of a splice recognition site in exon 4 and alternate polyA sites
- Exons 1-3 constitutive.
- Exon 4 used in thyroid cells.
- Exons 5 & 6 used in brain tissue.

- Exon choice (alternative splicing) is regulated by availability of a regulatory protein, SRp55, in thyroid tissue.

Alternative splicing expands the informational content of the genome

- At least 45-60 % of human genes produce primary transcripts that can be alternatively spliced to generate a great number of potential mRNAs and proteins.

- Up to 40% of genetic defects can be associated with incorrect RNA processing.

Take home message

Transcription of protein encoding genes is regulated at different levels in both prokaryotes and eukaryotes:

- Initiation of basal level of transcription (GTFs in eukaryotes, operons in prokaryotes).
- Enhancement of basal transcription (activators in both prokaryotes and eukaryotes).
- Tissue specificity (eukaryotes only-presence of activators limited to certain tissues).
- Processing of messenger RNA (eukaryotes-pre mRNA to mature mRNA through 5'cap, 3' poly A tail and removal of introns).